

ERIK TAYLOR, PhD

AI Systems Architect & ML Engineer | Agentic Systems, Dev Tooling & Research | Evaluation & Risk
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PhD engineer with 10+ years taking applied AI from research to shipped products. Architect of agentic LLM systems, the developer tooling that makes them buildable, and the evaluation harnesses that prove they work — most recently through agentic coding workflows. Prior: production computer vision at Sony, frontier-model evaluation at Snorkel AI, and >\$2M in NIH-funded computational imaging research, with 50+ peer-reviewed publications. A practice built on measurement, reliability, and safe deployment in regulated environments.

PROFESSIONAL EXPERIENCE

AI Research Lead & ML Engineer

Oct 2024 – Present

Taylor ML R&D | San Francisco, CA

- **Agentic AI Systems:** Architected apply-daemon (Apache-2.0, open-source) — an LLM-orchestrated agentic pipeline that triages roles through a multi-stage cascading-LLM workflow (extract → judge → evaluate), with human-in-the-loop review in Slack and a Claude Code CLI skill. Per-stage model routing, batch scoring, and a hard spend ceiling keep a daily run inside a fixed budget.
- **Developer Tooling for Agentic Engineering:** Designed and released dev-plan (MIT, open-source) — a Claude Code skill that converts planning discussions into scored, dependency-ordered development plans via a weighted implementation-priority model. Part of a broader skill suite covering cross-plan dependency coordination and structured ideation; published the design rationale in Digital Mind and ran a controlled comparison against the built-in planning baseline.
- **Agentic Development Practice:** Build and ship production systems with Claude Code (CLI and chat) for implementation, debugging, and iteration, backed by CI/CD, automated test suites, and lint gates on every push. Practiced across apply-daemon, dev-plan, and the autonomous portfolio manager.

AI Evaluation Specialist · Contract

Feb 2025 – Aug 2025

Snorkel AI | Remote

- **Frontier Model Risk & Reliability:** Authored advanced mathematical and graduate-level test cases designed to expose reasoning failure modes in frontier LLMs. Findings directly informed RL reward signals and alignment strategy for next-generation models and produced technical insights for production-readiness risk management.

Senior Algorithm Engineer

Jul 2022 – Oct 2024

Sony Electronics | San Jose, CA

- **Production Computer Vision at Scale:** Led end-to-end development of a two-stage vision pipeline shipped to phones and VR headsets — a sub-1 MB CNN with cascaded inference across sensor and host. Owned data curation, multi-GPU distributed training (DDP), INT8 quantization, and latency optimization (~30 FPS), securing a major B2B design win with a Fortune 500 customer.
- **Foundation-Model Roadmapping & Product Strategy:** Drove the technical roadmap for next-generation AI features by benchmarking SOTA vision and language foundation models — including a working prototype of open-vocabulary detection (OWL-ViT, CLIP) for promptable on-device perception — against performance, safety, and deployment constraints. Translated those trade-offs into product strategy.
- **Program Recovery:** Orchestrated a high-stakes response to a sudden data policy change by bridging legal, business, and academic partners to co-develop a compliant dataset. Unblocked a next-generation AI hardware initiative and rescued a multi-month delivery timeline for a flagship pre-market product.
- **Model Validation & Safety Criteria:** Defined evaluation protocols for on-device AI products targeting regulated markets, establishing internal standards for performance, bias, and deployment constraints aligned with responsible AI principles.

Research Assistant Professor, Medical Imaging AI

Apr 2019 – May 2022

University of New Mexico – Medical Physics & Radiology | Albuquerque, NM

- **Technical Program Leadership:** Led the AI strategy and computational infrastructure for a multi-disciplinary center with active NIH grant funding. Governed imaging data pipelines and model development across cardiac, neuro-oncology, and traumatic brain injury studies.
- **Biomedical Data Pipeline (Journal Cover Feature):** Rescued a stalled traumatic brain injury study by designing a custom deep learning framework (U-Net variant) capable of processing noisy, unstandardized biomedical data. Validated the framework against baselines, earning a cover feature in the Journal of Neurotrauma (2023).
- **Clinical Stakeholder Integration:** Bridged engineering researchers and clinical stakeholders (Radiology, Neurology, Oncology) to translate AI research into actionable clinical validation studies. Deployed HIPAA-compliant data handling, IRB-governed protocols, and clinical data governance across multi-site imaging studies.

NIH Research Fellow, Cardiovascular MRI

Jul 2016 – Mar 2019

Boston University | Boston, MA

- **Translational Imaging Research:** Developed and validated MRI acquisition and analysis methods for cardiovascular disease, combining physics-informed feature engineering with deep learning, and establishing a framework for choosing deep learning vs. classical methods by efficiency and clinical fit.

SELECTED PROJECTS

- **Autonomous Portfolio Manager (private).** +23.4% excess return vs SPY over 10 months of walk-forward paper trading (2025-09-22 → 2026-07-31; 2,008 model ratings → 181 positions; Sharpe 1.82, max drawdown -11.3%). Market-calibrated LLM agents reason over tools (financials, web search, filings retrieval) to generate investment theses; autonomous execution via Alpaca with risk guardrails and paper/live modes; 8-tab Streamlit dashboard for operator oversight. Evaluation platform with verified-reward decision tracking; real-cash mirror live since 2026-07. Comparative variants on LightGBM / TabPFN / XGBoost and agentic RAG. Full CI/CD on GitHub Actions; 602-test suite. Methodology and verified figures: eriktaylor.github.io/evidence/portfolio-manager
- **Digital Mind – Technical Series (Medium).** Author of a technical series on frontier AI systems – mathematical teardowns and worked implementations covering vision transformers, diffusion models, latent attention, and LLM reasoning and memorization.

TECHNICAL SKILLS

Agentic AI & LLM Applications: LLM application development (RAG, agents, tool use, function calling) · Claude / Claude Code (daily use) · Claude Code skill and plugin authoring · Anthropic API (via OpenRouter) · LangChain · LangGraph · LlamaIndex · autonomous agent orchestration · Model Context Protocol (MCP) · multi-modal AI · prompt engineering

Model Evaluation & AI Safety: Frontier-model reasoning failure analysis · evaluation harness design · model risk and validation · benchmarking (accuracy / latency / safety / cost trade-offs) · red-teaming · alignment-informed test design

Frameworks & Languages: Python (expert) · PyTorch · TensorFlow/Keras · Hugging Face Transformers · scikit-learn · LightGBM/XGBoost · FAISS · JavaScript · SQL · OpenCV

MLOps & Infrastructure: AWS · GCP · Docker · CI/CD (GitHub Actions) · Cloudflare (Pages, Workers) · distributed training (DDP, FSDP) · experiment tracking (W&B) · Linux/Ubuntu

Computer Vision: Detection, segmentation, depth estimation · open-vocabulary detection (OWL-ViT, CLIP) · self-supervised vision (DINOv2) · on-device / cascaded inference · quantization (INT8/INT4) · knowledge distillation · ONNX

Domain Knowledge: Medical imaging (MRI, PET, CT) · physics-informed ML · clinical data governance · translational research workflows · edge hardware constraints (memory, latency, power)

STRATEGIC CAPABILITIES

AI Product Strategy & Risk Management: Directing model evaluation and vendor selection · foundation model benchmarking · agentic and algorithmic stress-testing · defining model validation constraints · setting deployment and evaluation criteria across cloud and on-device targets.

Cross-Functional Program Leadership: Technical program management and full product lifecycle ownership · stakeholder management across clinical, legal, regulatory, and engineering teams · external partner and vendor coordination · matrixed team leadership.

EDUCATION & RESEARCH TRAINING

PhD, Biomedical Engineering – Brown University

Doctoral research in theranostic technologies, integrating advanced medical imaging diagnostics with targeted therapeutics.

NIH Research Fellow, Cardiovascular & Brain MRI – Boston University

Federally funded fellowship developing and evaluating deep learning frameworks for translational medical imaging.

BS, Biomedical Engineering – University of Texas at Austin

Graduate Coursework, Data Science – Harvard University

SELECTED PUBLICATIONS

Selected from 50+ peer-reviewed publications. Full list available on request.

Divani AA, Salazar P, Ikram HA, **Taylor E**, Wilson CM, et al. Non-invasive vagus nerve stimulation improves brain lesion volume and neurobehavioral outcomes in a rat model of traumatic brain injury. *Journal of Neurotrauma*. 2023; 40(13–14): 1481–1494. ★ **Journal Cover Feature**

Taylor EN, Huang N, Lin S, Mortazavi F, Wedeen VJ, et al. Lipid and smooth muscle architectural pathology in the rabbit atherosclerotic vessel wall using Q-space cardiovascular magnetic resonance. *Journal of Cardiovascular Magnetic Resonance*. 2022; 24(1): 74.

Taylor EN, Ding Y, Zhu S, Cheah E, et al. Association between tumor architecture derived from generalized Q-space MRI and survival in glioblastoma. *Oncotarget*. 2017; 8(26): 16296.